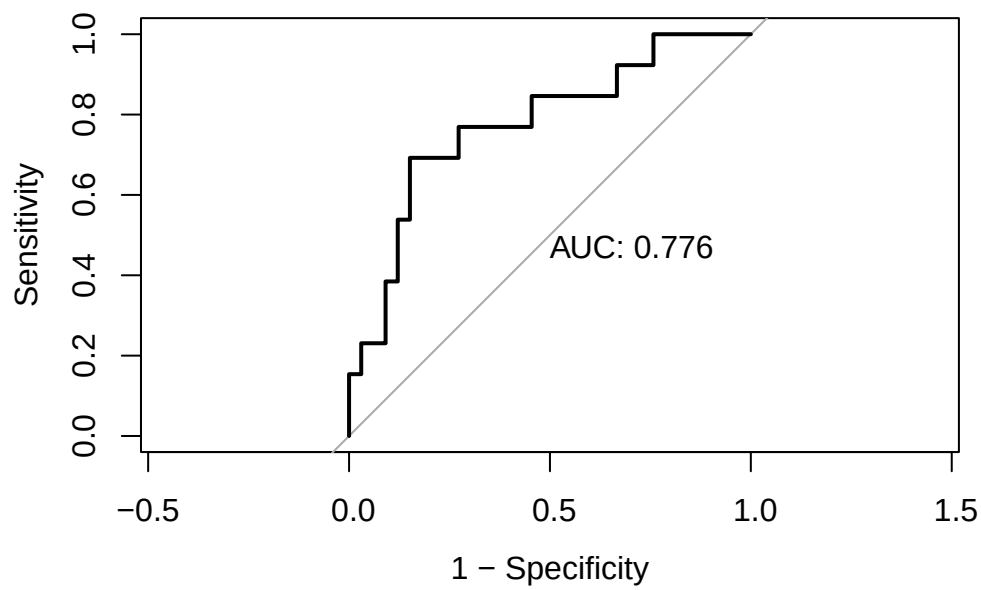


Final Project - 633A

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Introduction

Sephora is a well-known brand and storefront that specializes in beauty products across various categories such as hair, skincare, bath and body, and makeup. In this analysis, we are going to answer two research questions related to product performance and customer engagement. The first research question is to estimate the number of “loves” a product may receive online. “Love” count is essentially a measure of popularity where products with high engagement are estimated to have more “loves”. The ability to predict product popularity is crucial in maximizing revenue and optimizing marketing strategy. Our second research question is to predict whether a product will receive a good rating. Predicting whether a product will receive a high average rating can prevent unnecessary marketing spend on products that are unlikely to perform well or alternatively, identify well-performing products early on.

Data Description

We sourced the data file from Kaggle which contained 8,494 products and various information about the product such as number of reviews, whether it was an exclusive release, etc. The key variables used in this analysis include rating (average of values from 1-5), number of reviews, number of “loves”, price (USD), and two binary variables new and online only. After removing null values from the key variables, there was a clean dataset of 8,216 products. This dataset consisted of products from hundreds of brands across a number of product categories, as well as a wide range in the number of reviews. Initial attempts to meet linear model assumptions were a challenge thus in order to reduce the impact of outliers and extreme variability, we elected to only include products that were well-established, defined as having between 100 and 10,000 reviews and to focus on one brand. We decided to use the store’s own brand “Sephora Collection” which resulted in a final sample of 151 products and 13 variables.

Table 1: Summary statistics for Sephora Collection products

Characteristic	N = 151 ¹
Number of Loves	61,985 (73,322)
Number of Reviews	690 (1,040)
Mean Rating (1-5)	4.20 (0.52)
Product Price (USD)	16 (10)
Online Only	3 (2.0%)
Good Rating (≥ 4.5)	
Not Good	102 (68%)
Good	49 (32%)

Characteristic

N = 151¹

¹Mean (SD); n (%)

Shown in Table 1, Number of loves and number of reviews are highly variable, demonstrated by their large standard deviations when compared to their means[1]. The mean product rating is 4.2 (sd: 0.52) and the mean price is 16 (sd:10). Online only products seem to be rare for this brand, making up only 2% of products. Lastly, the good/not good rating split is about 30-70 indicating some inbalance the logistic regression model will have to consider.

Figure 1: Scatterplot matrix of key variables

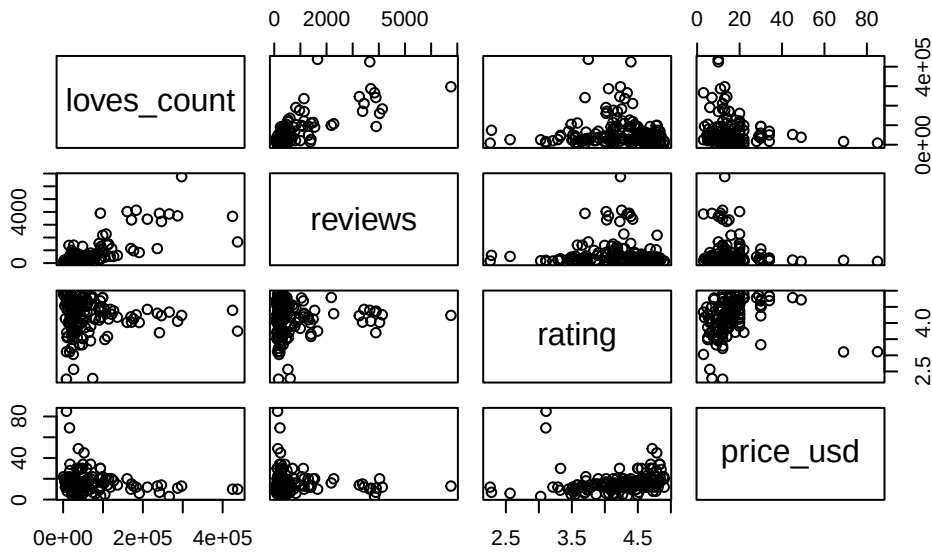


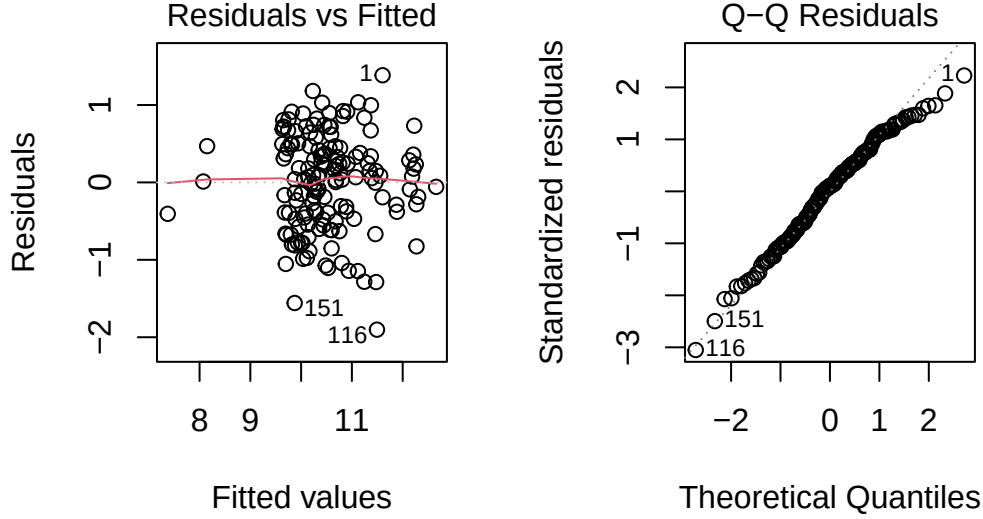
Figure 1 [2] confirms the variability and right-skewed data for both number of reviews and number of loves, suggesting both may be good candidates for log transformations. There also appears to be a potentially positive relationship between number of loves and rating. Lastly, the response variable “good rating” was created for the logistic regression model, which is defined as an average customer rating equal to or greater than 4.5.

Methods & Results

We first used a multiple linear regression model to evaluate popularity by using log transformed loves as the response variable. Polynomial terms, interactions and stepwise selection by Akaike Information Criterion (AIC) were used in order to finalize our model. When the final model was

reached it included rating, log transformed reviews, price, online-only, new and an interaction term[3]. Assumptions were reasonably met[4] including linearity and nonconstant variance as demonstrated by the roughly horizontal red line and no clear pattern in Figure 2. Normality of residuals was also roughly satisfied shown in Figure 3, with some expected deviation at the tails.

Figure 2 & 3: Residuals Plot and QQ Plot to Assess Linear Regression Assumptions



Multicollinearity was also assessed using variance inflation factors (VIF). The VIF values were close to the value of 1 indicating there was no evidence of multicollinearity. This suggests that the final multiple linear regression model is reliable. Our final multiple linear regression model was[5]:

```
log_loves ~ rating + log_reviews + new + online_only + rating:new
```

The results suggest that products with more reviews are associated with greater popularity, as measured by the number of loves. The new and online-only products also tend to have greater popularity. The effect of ratings on popularity depends on whether a product is new, as indicated by the interaction term. While higher ratings are associated with greater popularity, this relationship is not as strong for new products.

To examine the probability that a product receives a “good” rating, we used a logistic regression model. Using good rating as our response variable, we used log loves, price in USD and an interaction term between log transformed loves and price in USD. There was an interaction

term included to allow the effect of product popularity on the log-odds of a good rating to vary across different price levels[6].

The final model was chosen based on AIC score. We explored the use of splines but decided they did not improve the model compared to stepwise selection. The final model multiple logistic regression model was[7]:

```
good_rating ~ log_loves + price_usd + log_loves:price_usd
```

The logistic regression model suggests that the relationship between loves ($\log(\text{loves})$) and the log-odds of receiving a “good” rating depends on price. While a higher number of loves is associated with a lower log-odds of a good rating, this relationship becomes weaker as price increases. This shows that price influences how the number of log loves relates to ratings.

In order to assess model performance cross-validation was used. The model was trained on 70 percent of the data and tested on the remaining 30 percent. The performance of the model was assessed on accuracy, sensitivity, specificity and the area under the ROC curve. The results showed approximately 80 percent accuracy, 69 percent sensitivity and 85 percent specificity. The area under the ROC curve was .776 suggesting the model had moderate ability to distinguish between a “good” rating and those without.

Conclusion

This report examined the number of loves and the likelihood of receiving a high rating among Sephora Collection products. The multiple linear regression results showed that the number of loves is strongly associated with the number of reviews and weakly associated with customer ratings. New and online-only products tend to have more attention. The effect of ratings on loves is weaker for new products. This could mean that when a product is popular and new, it may not be because of ratings.

The factors that were focused on in the logistic regression model showed that the model had a moderate ability to predict whether a product received a “good” rating. The relationship between the number of loves and ratings was dependent on the price of the product. The number of loves tended to have a negative association with the probability of a good rating, although this relationship becomes weaker as price increases. This means that price plays a role in how customers react to products.

There were some limitations to this dataset, including that only Sephora Collection products were used, with a total of 151 products in the final dataset. There were only 3 online-only items, meaning that the results for online-only products may not be very reliable since there are so few in the dataset. The dataset was also messy and required transformations, which were crucial for the analysis but may make the results harder to interpret. Additionally, this analysis only applies to well-established products and may not generalize to new products.

Some recommendations for future work include using more dynamic statistical modeling methods, such as tree-based approaches, to better handle nonlinearity and variability. It would also be helpful to include more predictors in the model, such as text reviews or the number of purchases per product. Finally, including more brands and product categories would make the results more broadly applicable.

References

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning: With Applications in R (2nd ed.). Springer.

Data sourced from Kaggle: https://www.kaggle.com/datasets/melissamonfared/sephora-skincare-reviews?select=product_info_skincare.csv

Code Appendix

[1]

```
tbl_summary(  
  data = data2_sc  
  ,label = list(loves_count = "Number of Loves"  
                ,rating = "Mean Rating (1-5)"  
                ,reviews = "Number of Reviews"  
                ,price_usd = "Product Price (USD)"  
                ,online_only = "Online Only"  
                ,good_rating = "Good Rating (>=4.5)"  
                )  
  ,statistic = list(all_continuous() ~ "{mean} ({sd})"  
                    , all_categorical() ~ "{n} ({p}%)"  
                    )  
  ,include = c("loves_count", "reviews", "rating"  
               ,"price_usd", "online_only", "good_rating"  
               )  
  ) |>  
  as_flex_table()
```

[2]

```
pairs(loves_count ~ reviews + rating + price_usd, data = data2_sc)
```

[3]

```
summary(lm_step2)
```

Call:

```
lm(formula = log_loves ~ rating + log_reviews + new + online_only +  
    rating:new, data = data2_sc)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.90234	-0.40330	0.06681	0.44422	1.38436

Coefficients:

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

```

(Intercept)  5.94768    0.54263   10.961 < 2e-16 ***
rating       0.07844    0.10216    0.768 0.44381
log_reviews  0.72344    0.05230   13.833 < 2e-16 ***
new         24.89612    9.10093    2.736 0.00701 **
online_only  0.64786    0.37259    1.739 0.08419 .
rating:new   -5.59667    1.91162   -2.928 0.00397 **
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6293 on 145 degrees of freedom
Multiple R-squared: 0.6303, Adjusted R-squared: 0.6175
F-statistic: 49.43 on 5 and 145 DF, p-value: < 2.2e-16

[4]

```
plot(lm_step2, which = c(1,2))
```

[5]

```
formula(lm_step2)
```

[6]

```
summary(glm3)
```

Call:

```
glm(formula = good_rating ~ log_loves + price_usd + log_loves:price_usd,
     family = binomial, data = data2_sc)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	23.96827	5.81421	4.122	3.75e-05	***
log_loves	-2.55937	0.58841	-4.350	1.36e-05	***
price_usd	-0.91666	0.27623	-3.318	0.000905	***
log_loves:price_usd	0.09880	0.02815	3.511	0.000447	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 190.33 on 150 degrees of freedom
Residual deviance: 146.93 on 147 degrees of freedom
AIC: 154.93

Number of Fisher Scoring iterations: 5

[7]

```
formula(glm3)
```